AE 5335 Final Exam

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1. A space mission to be performed by an electrical rocket requires a velocity of 10000 m/s, a combined payload and vehicle structural mass of 1800 kg (excluding propellant and propulsion/engine system masses), and a propulsion/engine system mass (not including propellant mass) of 3200 kg. The accelerator efficiency is 55% and the power generation system delivers 150 Watts of power to the accelerator per kilogram of propulsion/engine system mass. Vehicle specific impulse is 2000 seconds. Accelerator efficiency is defined as in the notes (kinetic energy rate of exhaust over the electric power required by the accelerator).

   A. Find the propellant mass required for this mission
   B. Calculate the overall initial mass of the rocket

Characteristic velocity:

$$C = I_{sp} g_0 = 19,620\ m/s$$

The propellant mass is:

$$m_p = m_d \left(e^{\frac{\delta v}{C}} - 1 \right) = \boxed{3323.825\ kg}$$

The overall initial mass is:

$$m_0 = m_e + m_L + m_p = m_d + m_p = \boxed{8323.825\ kg}$$

2. An engineer in your group gives you the following specifications for a proposed space power generation system that he has designed. The system provides 1000 kWatts of power (provided from the power generation system for spacecraft propulsion and other power requirements). It utilizes a direct Brayton cycle with N_2 (nitrogen) as the heat engine working fluid. Emissivity of the radiator surface is 1.0. The average temperature of the N_2 during the heat addition segment of the Brayton cycle is 2000K. The average temperature of the N_2 as it flows through the radiator is 1000K. The radiator area required for this system is stated to be 8 m^2. Do you allow this design to go forward out of your group for development and what is the engineering basis for your decision?

For this problem I calculate the Carnot and Brayton efficiency.

The following is the Carnot efficiency:

$$\eta_c = 1 - \frac{T_L}{T_H} = 1 - \frac{T_{rad}}{T_{heat}} = 50\%$$

The following equation applies for any cycle:

$$\frac{A_{rad}}{W_{out}} = \frac{1}{\epsilon \sigma T_R^4}\left(\frac{1}{\eta} - 1\right)$$

Solving for the efficiency gives:

$$\eta_b = 69\%$$

This design should **not** move forward. The Carnot efficiency is less than that of the Brayton efficiency. This shows that there is a violation of the second law of thermodynamics.

3. In the year 2120, solar physicists determine that the sun will explode catastrophically in 50 years (with an uncertainty of plus or minus 16 years). This event will vaporize the inner planets (including Earth) and render the solar system uninhabitable. In addition, the shock wave from this event will travel outward in a spherical shell at enormous velocity (near light speed) such that any spacecraft located at a distance of less than 100 AU will also be utterly destroyed. (Let 1 AU = 93 million miles in this problem).

   A mission is immediately begun to design, build and launch *within four years* an interstellar survival ship (a sort of 'ark'). This ship will hopefully allow a small remnant of life on earth to escape this catastrophe. Its distant destination will be a star system in which a habitable planet has been detected.

   The spacecraft will be powered by an ion drive with a specific impulse of 3000 seconds. The power generation system delivers 200 Watts of power to the accelerator per kilogram of propulsion/engine system mass. Accelerator efficiency (see note on problem 1) is 50%. The payload, i.e., the mass of survivors plus supporting structure (not including propulsion/engine system mass or propellant mass), is estimated to be 2,000,000 kg.

   In order to make the construction of this rocket feasible with the time frame of four years, it will be designed as an **optimal electric rocket** (such that the overall mass that is required to be boosted and assembled in space is minimized).

   Based on the worst-case scenario, what must be the overall initial mass, the propellant mass, the engine mass, the mass flow rate of the propellant, the acceleration (firing) time, and the thrust required for this rocket, for it to 'just' avoid destruction?

   For this problem, ignore the acceleration time of the vehicle (but provide that information as requested above) in your calculation of the required ΔV. This is a 'free-space rocket' (no atmospheric drag and negligible gravity). You may also assume nearly instantaneous propagation of the shock wave through space, i.e., the vehicle had ***better*** be at the 'minimum survival distance' at the time the sun explodes.

The time left for the worst case is:

$$time = 50 - 16 - 4 = 30\ years = 946,080,000\ sec$$

The distance traveled in meters is:

$$distance = 1.49669 \times 10^{13}\ m$$

By ignoring the acceleration time the following is true:

$$V_{cruise} = \Delta V = \frac{distance}{time} = 15819.9\ m/s$$

Effective velocity:

$$C = I_{sp} g_0 = 28430\ m/s$$

For convenience:

$$\alpha = \frac{\Delta V}{C} = 0.5375$$

$$\beta = \beta_{opt} = \alpha \left(\frac{\alpha}{(2\,(e^{\alpha} - 1) - \alpha)} \right)^{1/2} = 0.4187$$

Find the burn time:

$$\tau_b = \left(\frac{\Delta V}{\beta} \right)^2 \frac{1}{2\alpha' \eta_a} = \boxed{82.619\ days}$$

The initial mass is:

$$m_0 = \frac{m_L}{(1 - (1 - e^{-\alpha})) \left[1 + \frac{\beta}{\alpha}^2 \right]} = \boxed{6,025,636\ kg}$$

The propellant mass is:

$$m_p = m_0 \left(1 - e^{\frac{\Delta V}{C}}\right) = \boxed{2,505,571\ kg}$$

The propellant mass flow rate is:

$$\dot{m_p} = \frac{m_p}{\tau_b} = \boxed{0.351\ kg/s}$$

The optimal engine mass is:

$$\dot{m_e} = m_0 - \left(m_L - m_p\right) = \boxed{1,520,065\ kg}$$

The thrust required is:

$$Thrust = \dot{m_p} C = \boxed{10,330\ N}$$

4. A low-altitude fixed area non-afterburning turbojet with on-design compressor ratio of 20 is flown on-design at $M_0 = 2$, $T_0 = 300K$ and $P_0 = 101325$ N/m^2 where it produces 50000 N of thrust. The maximum temperature in the engine is 1500K.

   Calculate the thrust obtained by this engine when it is flown at an off-design Mach number of 1.0 with the same T_0 and P_0 (same altitude) and with the same maximum temperature in the engine. Assume that the nozzle remains fully expanded at both flight Mach numbers, i.e., $P_9 = P_0$. Use ratio of specific heats = 1.4 and R = 287 and assume that the engine is ideal including that fuel air ratio is small with respect to 1.0.

On Design:

$M_{0R} = 2, T_{0R} = 300K, P_{0R} = 101325\ Pa, P_{9R} = P_{0R}, T_{t4R} = 1500\ K, \pi_{cR} = 20$

$$u_{0R} = M_{0R}\sqrt{\gamma R T_{0R}} = 694.377\ m/s$$

$$\tau_{rR} = 1 + \frac{\gamma - 1}{2} M_{0R}^2 = 1.8$$

$$\pi_{rR} = \tau_{rR}^{\frac{\gamma}{\gamma-1}} = 7.824$$

$$\tau_{cR} = \pi_{cR}^{\frac{\gamma-1}{\gamma}} = 2.354$$

$$\tau_{\lambda R} = \frac{T_{t4R}}{T_{0R}} = 5$$

$$\tau_{bR} = \frac{\tau_{\lambda R}}{\tau_{rR}\tau_{cR}} = 1.180$$

$$\tau_{tR} = 1 - \frac{\tau_{rR}}{\tau_{\lambda R}(\tau_{cR} - 1)} = 0.513$$

$$\pi_{tR} = \tau_{tR}^{\frac{\gamma}{\gamma-1}} = 0.097$$

$$\frac{P_{t9R}}{P_{9R}} = \pi_{tR}\tau_{bR}\pi_{cR} = 15.103$$

$$M_{9R} = \frac{\left(\frac{P_{t9R}}{P_{9R}}\right)^{\frac{\gamma-1}{\gamma}} - 1}{\left(\frac{\gamma-1}{2}\right)^{1/2}} = 2.421$$

$$T_{t9R} = T_{0R}\tau_{tR}\tau_{bR}\tau_{cR}\tau_{rR} = 769.085\ K$$

$$T_{9R} = \frac{T_{t9R}}{1 + \frac{\gamma-1}{2} M_{9R}^2} = 354.076\ K$$

$$u_{9R} = M_{9R}\sqrt{\gamma R T_{9R}} = 913.101\ m/s$$

$$\dot{m}_{airR} = \frac{F_R}{u_{9R} - u_{0R}} = 228.599\ kg/s$$

$$A_{4R} = \dot{m}_{airR}\frac{\sqrt{R}}{\Gamma P_{0R}\pi_{rR} pi_{cR}}\sqrt{\tau_{\lambda R} T_{0R}} = 0.01381\ m^2$$

Off Design:

Using Fat Jet Assumptions:

$M_0 = 1, T_0 = T_{0R}, P_0 = P_{0R}, P_9 = P_0, \tau_t = \tau_{tR}, T_{t4} = T_{t4R}, \pi_t = \pi_{tR}, \tau_\lambda = 5$

$$u_0 = M_0\sqrt{\gamma R T_0} = 347.189\ m/s$$

$$\tau_r = 1 + \frac{\gamma - 1}{2} M_0^2 = 1.2$$

$$\pi_r = \tau_r^{\frac{\gamma}{\gamma-1}} = 1.893$$

$$\tau_c = 1 + \frac{\tau_\lambda}{\tau_r}(1 - \tau_t) = 3.03$$

$$\tau_b = \frac{\tau_\lambda}{\tau_r \tau_c} = 1.375$$

$$\pi_{cR} = \tau_{cR}^{\frac{\gamma}{\gamma-1}} = 48.441$$

$$\dot{m}_{airR} = \frac{A_4}{\frac{\sqrt{R}}{\Gamma P_0 \pi_r p i_c}\sqrt{\tau_\lambda T_0}} = 133.948\ kg/s$$

$$\frac{P_{t9}}{P_9} = \pi_t \tau_b \pi_c = 8.850$$

$$M_9 = \frac{\left(\frac{P_{t9}}{P_9}\right)^{\frac{\gamma-1}{\gamma}} - 1}{\left(\frac{\gamma-1}{2}\right)^{1/2}} = 2.079$$

$$T_{t9} = T_0 \tau_t \tau_b \tau_c \tau_r = 769.085\ K$$

$$T_9 = \frac{T_{t9}}{1 + \frac{\gamma-1}{2} M_9^2} = 412.498\ K$$

$$u_9 = M_9\sqrt{\gamma R T_9} = 846.394\ m/s$$

$$F = \dot{m}_{air}(u_9 - u_0) = \boxed{66867.392\ N}$$

5. Provide the preliminary design for a compressor blade, based on free-vortex theory; specifically provide well-labeled plots of leading and trailing edge angles of both the rotor and stator blades from hub to tip, as well as well-labeled plots of the degree of reaction for the stage, the diffusion factor for rotor, and the diffusion factor for stator, all from hub to tip. Identify potential blade locations where fluid separation is likely. See below for additional information regarding requirements.

   This is a 14-stage compressor (repeating stages) and the total pressure rise across each stage is 35000 N/m^2. RPM of the spool is 30000. The stage of interest in this problem is an intermediate stage where the tip radius for the turbomachinery is 0.1086 m and the hub radius is 0.03676 m. The jet engine is an annular (axi-symmetric) design. Axial velocity in this stage is 130.0 m/s and static density (entering) is 1.32 kg/m^3. Mandate the degree of reaction of 0.5 and a blade solidity of 1.0 at the mass-averaged radius.

   Finally (for small but finite credit), if the chord of both rotor and stator blades at the mass averaged radius is 0.020376 m, provide the number of rotor blades and the number of stator blades for this stage.

   Your results should consist of a plot of leading and trailing edge angles for the rotor (single plot), a plot of leading and trailing edge angles for the stator (single plot), a plot of degree of reaction for the stage (single plot), and a plot of diffusion factors for the rotor and for the stator (single plot), for a total of four plots. Also include a brief narrative identification of the regions where separation on the blades may be an issue and give the number of blades in each blade row within this stage.

   Provide relevant description of methodology, equations solved, etc. after these results.

Radius values spaced linearly between the hub and tip radius:

$$r = linspace(r_h, r_t, N_{steps})$$

Degree of reaction at a given location:

$$^\circ R = 1 - \frac{r_m^2}{r^2}\,(1 -^\circ R_m)$$

Where:

$$r_m = \sqrt{\frac{r_t^2 + r_h^2}{2}}$$

Degree of reaction at the hub:

$$^\circ R_h =^\circ R[0]$$

Swirl velocity equations:

$$V_{\theta_1 h} = \frac{w_0}{2}\left[\frac{2\omega r_h}{w_0}\,(1 -^\circ R_h) - \frac{1}{\frac{\omega r_h}{w_0}}\frac{\Delta P_t}{\rho_0 w_0^2}\right]$$

$$V_{\theta_2 h} = \frac{w_0}{2}\left[\frac{2\omega r_h}{w_0}\,(1 -^\circ R_h) + \frac{1}{\frac{\omega r_h}{w_0}}\frac{\Delta P_t}{\rho_0 w_0^2}\right]$$

$$V_{\theta_1} = \frac{r_h V_{\theta_1 h}}{r}$$

$$V_{\theta_2} = \frac{r_h V_{\theta_2 h}}{r}$$

Equations for stator leading and trailing edge angles

$$\theta_1 = \tan^{-1}\frac{V_{\theta_1}}{w_0}$$

$$\theta_2 = \tan^{-1}\frac{V_{\theta_2}}{w_0}$$

Equations for rotor leading and trailing edge angles

$$\beta_1 = \tan^{-1}\left(\frac{\omega r}{w_0} - \frac{V_{\theta_1}}{w_0}\right)$$

$$\beta_2 = \tan^{-1}\left(\frac{\omega r}{w_0} - \frac{V_{\theta_2}}{w_0}\right)$$

Equation for rotor diffusion:

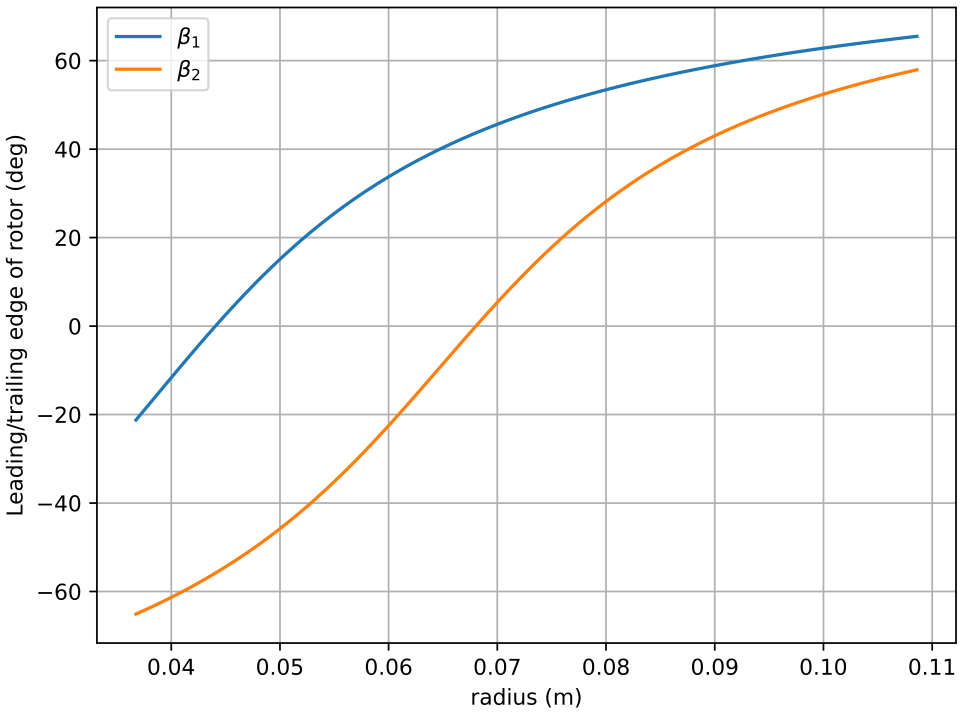
$$D_{rotor} = 1 - \frac{\cos(\beta_1)}{\cos(\beta_2)} + \frac{|\tan\beta_1 - \tan\beta_2|}{2\sigma\cos\beta_1}$$

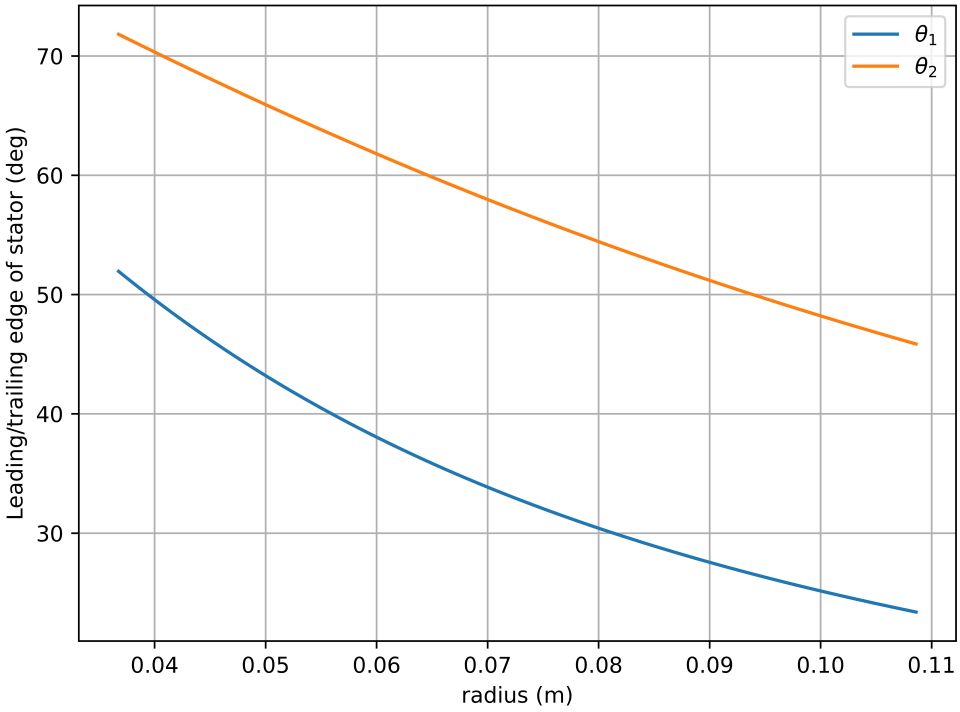
Equation for stator diffusion:

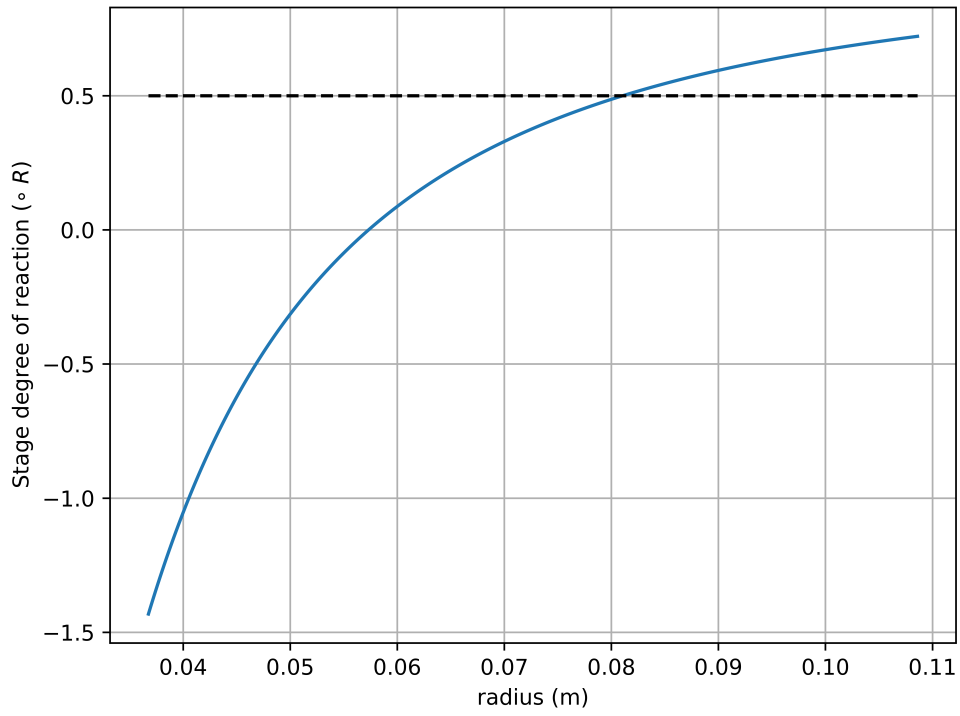
$$D_{stator} = 1 - \frac{\cos(\theta_2)}{\cos(\theta_3)} + \frac{|\tan\theta_3 - \tan\theta_2|}{2\sigma\sec\theta_2}$$

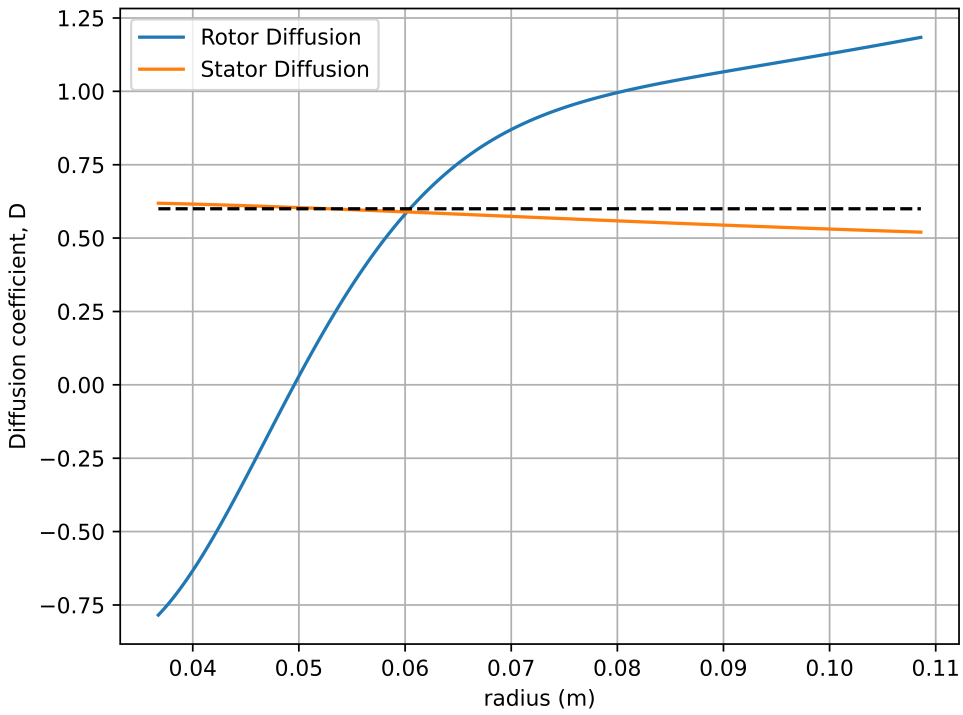
Where:

$$\sigma = \frac{r_m}{r}\sigma_m$$




θ_1
θ_2
Leading/trailing edge of stator (deg)
70
60
50
40
30
0.04
0.05
0.06
0.07
0.08
0.09
0.10
0.11
radius (m)


Stage degree of reaction ($\circ$ R)
0.5
0.0
−0.5
−1.0
−1.5
0.04
0.05
0.06
0.07
0.08
0.09
0.10
0.11
radius (m)



For the stator separation can occur at the hub and for the rotor separation will likely occur at $\boxed{r = 0.06}$ **meters**

There are 25 blades for the row in which the chord is given. This was solved by using the following equation:

$$\boxed{blades = \frac{2\pi r_m}{chord} = 25}$$

```
import numpy as np
import matplotlib.pyplot as plt

stages = 14
dP = 35_000
RPM = 30000
omega = RPM/30 * np.pi

r_tip = 0.1086
r_hub = 0.03676
w0 = 130
rho = 1.32

doR = 0.5

sigma_m = 1.0 #solidity at mass average radius

itr = 1_000
r = np.linspace(r_hub,r_tip,itr)
rm = np.sqrt((r_tip**2 + r_hub**2)/2)

R = 1 - rm**2 / r**2 * (1 - doR)
Rh = R[0]

V_theta_1_h = w0* 1/2 * (2*omega*r_hub/w0*(1 - Rh) - 1/(omega*r_hub/w0)*(dP/(rho*w0**2)))
V_theta_2_h = w0* 1/2 * (2*omega*r_hub/w0*(1 - Rh) + 1/(omega*r_hub/w0)*(dP/(rho*w0**2)))


V_theta_1 = r_hub * V_theta_1_h / r
V_theta_2 = r_hub * V_theta_2_h / r

theta_1 = np.arctan(V_theta_1/w0)
theta_2 = np.arctan(V_theta_2/w0)

beta_1 = np.arctan(omega*r/w0 - V_theta_1/w0)
beta_2 = np.arctan(omega*r/w0 - V_theta_2/w0)

sigma = rm / r * sigma_m

D_rotor = 1 - np.cos(beta_1)/np.cos(beta_2) + 1/(2*sigma) * np.abs(np.tan(beta_1)-np.tan(beta_2))
                                                *1 /np.cos(beta_1)

theta_3 = theta_1
D_stator = 1 - np.cos(theta_2)/np.cos(theta_3) + np.abs(np.tan(theta_3) - np.tan(theta_2))/(2*
                                                sigma*1/np.cos(theta_2))

fig = plt.figure(1)
ax = plt.axes()
ax.plot(r,np.rad2deg(theta_1),label = r'$\theta_1$')
ax.plot(r,np.rad2deg(theta_2),label = r'$\theta_2$')


ax.set_ylabel('Leading/trailing edge of stator (deg)')
ax.set_xlabel(r'radius (m)')
plt.tight_layout()
ax.grid()
ax.legend()
plt.savefig(r'stator_angles.png', dpi=1200)

plt.show()

fig = plt.figure(2)
ax = plt.axes()
ax.plot(r,np.rad2deg(beta_1),label = r'$\beta_1$')
ax.plot(r,np.rad2deg(beta_2),label = r'$\beta_2$')

ax.set_ylabel('Leading/trailing edge of rotor (deg)')
```

```
ax.set_xlabel(r'radius (m)')
plt.tight_layout()
ax.legend()
ax.grid()
plt.savefig(r'rotor_angles.png', dpi=1200)
plt.show()

fig = plt.figure(3)


ax = plt.axes()
ax.plot(r,R)
line = np.ones(itr) * doR
ax.plot(r,line,ls='--',color = 'k')


ax.set_ylabel(r'Stage degree of reaction $ \left(\circ^ R \right)$')
ax.set_xlabel(r'radius (m)')
plt.grid()
plt.tight_layout()
plt.savefig(r'dor.png', dpi=1200)
plt.show()
fig = plt.figure(4)
ax = plt.axes()
ax.plot(r,D_rotor, label = "Rotor Diffusion")
ax.plot(r,D_stator,label = "Stator Diffusion")
line = np.ones(itr) * 0.6
ax.plot(r,line,ls='--',color = 'k')


ax.set_ylabel(r'Diffusion coefficient, D')
ax.set_xlabel(r'radius (m)')
plt.grid()
plt.tight_layout()
plt.legend()
plt.savefig(r'diffusion.png', dpi=1200)
plt.show()

chord = 0.020376 # m
Cm = rm * 2*np.pi
blades = Cm / chord
print(f'{blades = }')
```